

Electrostatic Problem in Bipolar Coordinates

1 Statement of the Problem

The electrostatic problem of two infinitely long, parallel cylindrical conductors has been studied by both Jackson [1] and Landau [2], using the method of image charges. While effective, this approach can appear somewhat *ad hoc*, since its validity is not justified *a priori*. In this note, we instead solve the problem using a more systematic and traditional approach by directly solving Laplace equation in bipolar coordinates with appropriate boundary conditions. We also construct the corresponding Green function, following the procedure outlined in Jackson's text.

2 Bipolar Coordinates

From our experience with a single conducting cylinder or sphere, we know that if we can find a coordinate system in which a coordinate surface coincides exactly with the conducting surface under consideration, the electrostatic boundary-value problem becomes greatly simplified. In such cases the boundary conditions can be imposed directly on surfaces of constant coordinate, and Laplace equation often admits separable solutions.

For the present problem of two parallel cylindrical conductors, the most natural coordinate system is provided by bipolar coordinates. This system is particularly well suited to geometries involving two circles (or, in three dimensions, two infinitely long cylinders), since surfaces of constant coordinate correspond precisely to such shapes. Before introducing bipolar coordinates explicitly, it is instructive to consider the geometry of the Apollonius circles, which underlies their construction.

2.1 Apollonius Circles

A circle is usually defined as the set of points at a fixed distance r (the radius) from a given point, the center of the circle. However, there exist other equivalent geometric characterizations. In particular, Apollonius showed that a circle can also be defined as the locus of points whose distances to two fixed points have a constant ratio. These two fixed points are known as the foci.

Let the two foci be separated by a distance $2a$ and located at $(\pm a, 0)$ on the x -axis. The locus of points whose distances to the two foci have a constant ratio C therefore satisfies

$$\frac{\sqrt{(x+a)^2 + y^2}}{\sqrt{(x-a)^2 + y^2}} = C.$$

Squaring both sides and rearranging terms, we will obtain

$$(1 - C^2)x^2 + 2(1 + C^2)ax + (1 - C^2)a^2 + (1 - C^2)y^2 = 0,$$

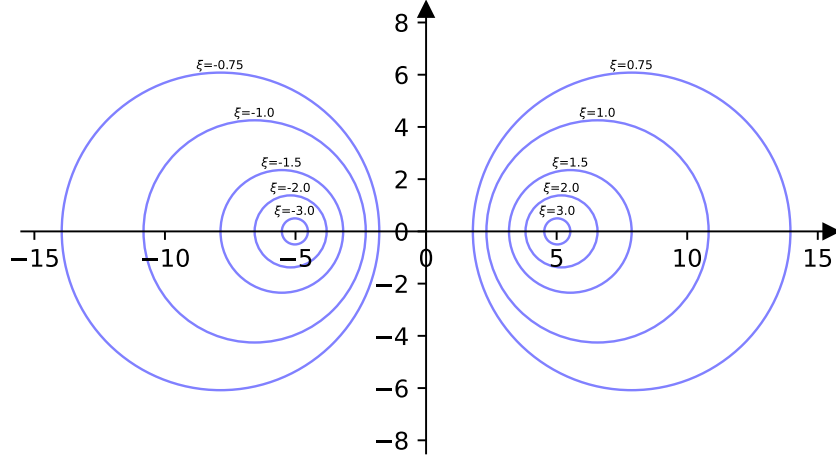


Figure 1: The first family of Apollonius circles, with $a = 5$.

which can be rewritten in a more familiar form,

$$\left(x - \frac{C^2 + 1}{C^2 - 1}a\right)^2 + y^2 = \frac{4C^2}{(C^2 - 1)^2}a^2,$$

which can be recognized as an equation of circle with center $(x_c, 0)$ and radius given by

$$x_c = \frac{C^2 + 1}{C^2 - 1}a, \quad R = \frac{2C}{|C^2 - 1|}a,$$

respectively. The circles obtained by varying the constant C are known as the first family of Apollonius circles (see Fig. 1). For later convenience, let $C = e^\xi$, then we will have

$$x_c = \frac{a}{\tanh \xi}, \quad R = \frac{a}{\sinh |\xi|}.$$

Clearly, circles whose centers lie in the right half-plane correspond to $\xi > 0$, while those with centers in the left half-plane correspond to $\xi < 0$. On the y -axis, we have $\xi = 0$.

In addition to the first family of Apollonius circles, there is a second family of circles that are orthogonal to those in the first family. These circles have their centers on the y -axis and are described by the equation

$$x^2 + (y - y_c)^2 = R^2,$$

with

$$y_c = \frac{a}{\tan \eta}, \quad R = \frac{a}{\sin |\eta|}, \tag{1}$$

and $\eta \in [0, \pi]$ (see Fig. 2). All circles in this family pass through the two foci. For $\eta = \pi$, the circle degenerates to a line segment connecting the two foci, whereas for $\eta = 0$, it corresponds to the two rays

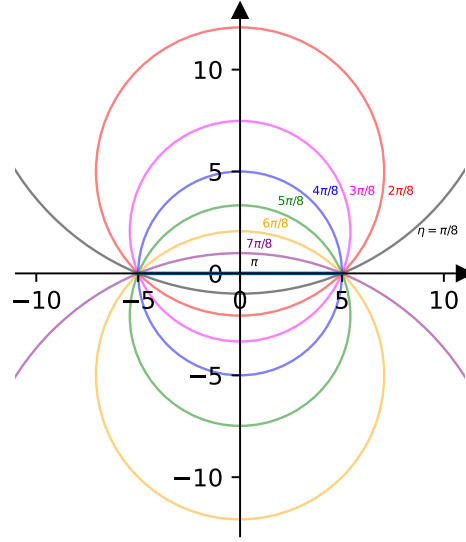


Figure 2: The second family of Apollonius circles.

extending along the x -axis with $|x| > a$. According to the inscribed angle theorem, the parameter η can be identified as the inscribed angle, and it is constant along each circle in the second family. As illustrated in Fig. 3, $\eta = \theta_1 - \theta_2$, with θ_1 and θ_2 are the angles formed between the lines connecting a point on the circle to the two foci and the x -axis. Using the geometric relations shown in Fig. 3, Eq. (1), which gives the center and the radius of the circles of the second family, can be derived.

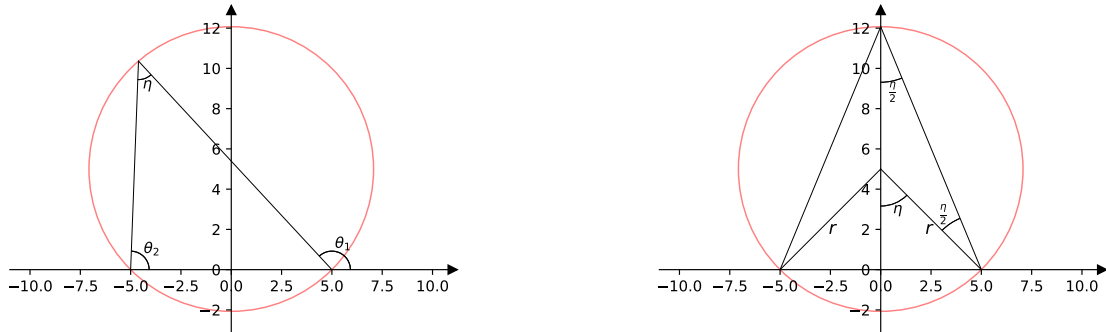


Figure 3: The second family of Apollonius circles.

As illustrated in Fig. 4, each circle from the first family intersects every circle from the second family at two points, one located above the x -axis and the other below it. Consequently, assigning the coordinates (ξ, η) to a point on the plane can be ambiguous. To remove this ambiguity, we can extend the range of the parameter η from $[0, \pi]$ to $[0, 2\pi]$. Due to the periodicity of the sine and tangent

functions, circles with η values differing by π share the same center and radius. To distinguish points above and below the x -axis, we assign $\eta \in [0, \pi]$ to points above the axis and $\eta \in [\pi, 2\pi]$ to points below the axis (see Fig. 5).

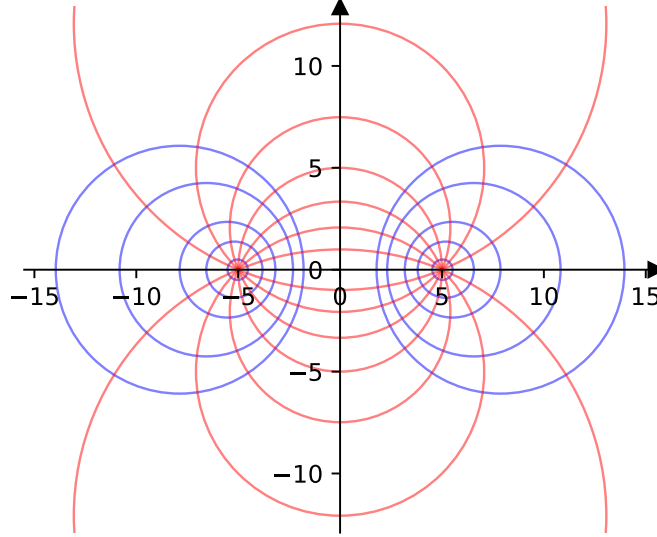


Figure 4: Apollonius circles of the first and the second families.

2.2 Bipolar Coordinates from Conformal Mapping

In the previous section, we described the two families of Apollonius circles, which form a basis for an orthogonal coordinate system. To obtain explicit formulas for the bipolar coordinates, we can exploit the fact that orthogonal coordinates can be generated via conformal mapping. Specifically, by applying a holomorphic function $f(z)$ to a standard 2D Cartesian grid, where $z = x + iy$, we map points to a new complex plane $w = \xi + i\eta$. In fact, to align with the sign of the inscribed angle, we will write $w = \xi - i\eta$.

Since

$$\xi = \frac{1}{2} \log \left(\frac{(x+a)^2 + y^2}{(x-a)^2 + y^2} \right),$$

we can identify it as the real part of the function

$$f(z) = \log \left(\frac{z+a}{z-a} \right),$$

and

$$\xi = \operatorname{Re} \left[\log \left(\frac{z+a}{z-a} \right) \right].$$

Then,

$$\eta = -\operatorname{Im} \left[\log \left(\frac{z+a}{z-a} \right) \right] = \operatorname{Arg}(z-a) - \operatorname{Arg}(z+a),$$

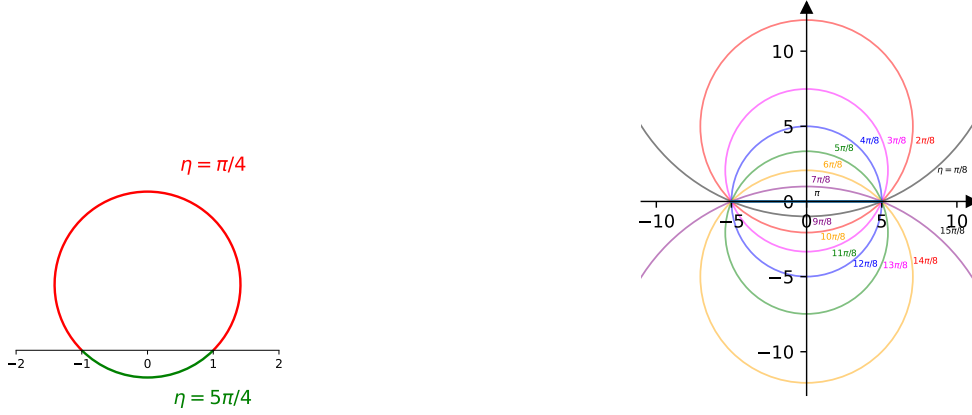


Figure 5: Parameter η for the second family of circles. On the same circle, points above and below the x -axis are labelled by η values differing by π .

which is exactly the difference in the angles formed between the lines connecting a point on the circle to the two foci and the x -axis.

To obtain the explicit bipolar coordinate transformation, notice that

$$w = e^{\xi - i\eta} = \frac{z + a}{z - a},$$

we will have

$$\begin{aligned} \frac{z}{a} &= \frac{w + 1}{w - 1} = \frac{e^{\xi - i\eta} + 1}{e^{\xi - i\eta} - 1} = \frac{e^{\xi} \cos \eta + 1 - ie^{\xi} \sin \eta}{e^{\xi} \cos \eta - 1 - ie^{\xi} \sin \eta} \\ &= \frac{(e^{\xi} \cos \eta + 1 - ie^{\xi} \sin \eta)(e^{\xi} \cos \eta - 1 + ie^{\xi} \sin \eta)}{(e^{\xi} \cos \eta - 1)^2 + e^{2\xi} \sin^2 \eta} \\ &= \frac{e^{2\xi} - 1 + 2ie^{\xi} \sin \eta}{e^{2\xi} - 2e^{\xi} \cos \eta + 1} = \frac{e^{\xi} - e^{-\xi} + 2i \sin \eta}{e^{\xi} - 2 \cos \eta + e^{-\xi}} \\ &= \frac{\sinh \xi + i \sin \eta}{\cosh \xi - \cos \eta}. \end{aligned}$$

Therefore,

$$x = \frac{a \sinh \xi}{\cosh \xi - \cos \eta}, \quad y = \frac{a \sin \eta}{\cosh \xi - \cos \eta}. \quad (2)$$

2.3 Scale Factors, Unit Vectors, and the Laplacian Operator

Let $\mathbf{r} = (x, y)$, then by Eq. (2), we have

$$\frac{\partial \mathbf{r}}{\partial \xi} = \left(\frac{\partial x}{\partial \xi}, \frac{\partial y}{\partial \xi} \right) = \frac{a}{(\cosh \xi - \cos \eta)^2} (1 - \cosh \xi \cos \eta, -\sinh \xi \sin \eta),$$

and

$$\frac{\partial \mathbf{r}}{\partial \eta} = \left(\frac{\partial x}{\partial \eta}, \frac{\partial y}{\partial \eta} \right) = \frac{a}{(\cosh \xi - \cos \eta)^2} (-\sinh \xi \sin \eta, \cosh \xi \cos \eta - 1),$$

which lead to

$$\left| \frac{\partial \mathbf{r}}{\partial \xi} \right|^2 = \left| \frac{\partial \mathbf{r}}{\partial \eta} \right|^2 = \frac{a^2}{(\cosh \xi - \cos \eta)^2}.$$

Then, we can find that the scale factors h_ξ and h_η are equal,

$$h_\xi = h_\eta = \frac{a}{\cosh \xi - \cos \eta}.$$

The unit vectors in the bipolar coordinates can be determined as

$$\hat{e}_\xi = \frac{1}{h_\xi} \frac{\partial \mathbf{r}}{\partial \xi} = \frac{1}{\cosh \xi - \cos \eta} (1 - \cosh \xi \cos \eta, -\sinh \xi \sin \eta),$$

and

$$\hat{e}_\eta = \frac{1}{h_\eta} \frac{\partial \mathbf{r}}{\partial \eta} = \frac{1}{\cosh \xi - \cos \eta} (-\sinh \xi \sin \eta, \cosh \xi \cos \eta - 1),$$

which are different at each point on the plane (see Fig. 6). Finally, the Laplace operator is given by

$$\nabla^2 f = \frac{(\cosh \xi - \cos \eta)^2}{a^2} \left(\frac{\partial^2 f}{\partial \xi^2} + \frac{\partial^2 f}{\partial \eta^2} \right).$$

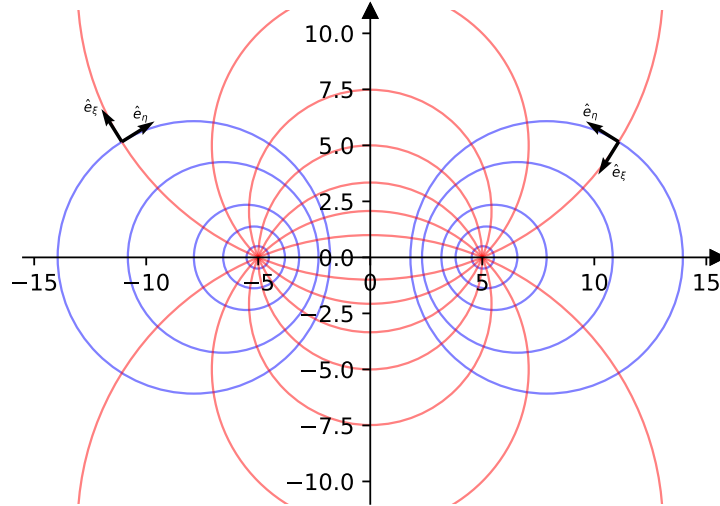


Figure 6: Unit vectors in the bipolar coordinate system.

2.4 Bipolar Coordinates from Two Nonconcentric Circles

So far, we have considered only the case in which the locations of the two foci are given. In this section, we extend the construction by deriving bipolar coordinates from two nonconcentric circles, without assuming that the foci are known. This approach allows the coordinate system to be defined

directly from the geometry of the circles, providing a more general and flexible framework for problems in electrostatics and potential theory.

Consider two circles of radii R_1 and R_2 , separated by a distance of d . Assume that the corresponding foci lie on the x -axis at $(\pm a, 0)$. In bipolar coordinates, the radii of the circles are related to the ξ parameters of their respective coordinates surfaces by

$$R_2 = \frac{a}{\sinh \xi_2}, \quad R_1 = \phi_1 \frac{a}{\sinh \xi_1}, \quad d = \frac{a}{\tanh \xi_2} - \frac{a}{\tanh \xi_1}.$$

Here, we assume that ξ_2 is always positive, while ξ_1 may be either positive or negative. We introduce a sign parameter $\phi_1 = \pm 1$ to indicate whether the circle of radius R_1 lies on the same side of the plane as the other circle ($\phi_1 = +1$) or on the opposite side ($\phi_1 = -1$). To determine the bipolar coordinates, we must solve three unknowns ξ_1 , ξ_2 , and a using the above three equations. Consequently, a unique bipolar coordinate system can be established from exactly two nonconcentric circles; adding more circles would overconstrain the system and generally make a consistent solution impossible.

Since

$$\frac{1}{\tanh \xi} = \text{sgn}(\xi) \sqrt{1 + \frac{1}{\sinh^2 \xi}} = \text{sgn}(\xi) \sqrt{1 + \frac{R^2}{a^2}},$$

the distance d between the centers of the two circles can be expressed as

$$d = a \left(\sqrt{1 + \frac{R_2^2}{a^2}} - \phi_1 \sqrt{1 + \frac{R_1^2}{a^2}} \right) = \sqrt{R_2^2 + a^2} - \phi_1 \sqrt{R_1^2 + a^2}.$$

Solving this equation, we will get

$$a^2 = \frac{1}{4d^2} [(R_1 + R_2)^2 - d^2] [(R_1 - R_2)^2 - d^2].$$

For $\phi_1 = 1$, the two circles lie on the same side of the plane, with the circle of larger ξ enclosed by the other. In this case, the geometric constraints require $R_1 + R_2 > d$ and $|R_1 - R_2| > d$, and

$$a = \frac{1}{2d} [(R_1 + R_2)^2 - d^2]^{1/2} [(R_1 - R_2)^2 - d^2]^{1/2}.$$

On the other hand, for $\phi_1 = -1$, the two circles lie on opposite sides of the plane, with $R_1 + R_2 < d$ and $|R_1 - R_2| < d$, and

$$a = \frac{1}{2d} [d^2 - (R_1 + R_2)^2]^{1/2} [d^2 - (R_1 - R_2)^2]^{1/2}$$

Once a is determined, the ξ parameters for the two circles can be obtained from the relations

$$\xi_1 = \phi_1 \text{arcsinh} \left(\frac{a}{R_1} \right), \quad \xi_2 = \text{arcsinh} \left(\frac{a}{R_2} \right).$$

Now, the bipolar coordinate system is completely determined from the two nonconcentric circles.

3 Electrostatic Problem in Bipolar Coordinates

Consider two infinitely long, parallel cylindrical conductors of radii R_1 and R_2 , separated by a distance of d . Our goal is to determine the capacitance per unit length between the cylinders. To do this, we must solve the two-dimensional Laplace equation,

$$\nabla^2 \Phi = 0,$$

where $\Phi(x, y)$ is the electrostatic potential in the plane perpendicular to the axes of the cylinders. In bipolar coordinates,

$$\nabla^2 \Phi = \frac{(\cosh \xi - \cos \eta)^2}{a^2} \left(\frac{\partial^2 \Phi}{\partial \xi^2} + \frac{\partial^2 \Phi}{\partial \eta^2} \right),$$

so the Laplace equation reduces to

$$\frac{\partial^2 \Phi}{\partial \xi^2} + \frac{\partial^2 \Phi}{\partial \eta^2} = 0.$$

Remarkably, this is formally identical to the Laplace equation in Cartesian coordinates, which allows us to use standard separation-of-variables techniques to solve for $\Phi(\xi, \eta)$. The general solution to the reduced Laplace equation can be written as

$$\Phi(\xi, \eta) = A_0 + B_0 \xi + \sum_{k \neq 0} (A_k e^{-k\xi} + B_k e^{k\xi}) e^{ik\eta},$$

where A_0, B_0, A_k, B_k are constants to be determined from the boundary conditions on the cylindrical surfaces.

Imposing the boundary conditions that the cylindrical conductors are equipotential on their surfaces, we have

$$\Phi(\xi_1, \eta) = V_1, \quad \Phi(\xi_2, \eta) = V_2,$$

where V_1 and V_2 are the potentials of the two cylinders, and ξ_1 and ξ_2 correspond to their respective surfaces in bipolar coordinates. Let $V = V_2 - V_1$, the coefficients in the general solution are then determined as

$$A_0 = \frac{\xi_2 V_1 - \xi_1 V_2}{\xi_2 - \xi_1}, \quad B_0 = \frac{V}{\xi_2 - \xi_1},$$

while all other coefficients vanish, $A_k = B_k = 0$ for $k \neq 0$. With these values, the potential depends only on ξ and varies linearly between the surfaces of the two cylinders,

$$\Phi(\xi, \eta) = \frac{(\xi_2 - \xi)V_1 + (\xi - \xi_1)V_2}{\xi_2 - \xi_1}.$$

From the electrostatic potential Φ , the electric field is its gradient,

$$\mathbf{E} = -\nabla \Phi = -\frac{1}{h_\xi} \frac{\partial \Phi}{\partial \xi} \hat{e}_\xi - \frac{1}{h_\eta} \frac{\partial \Phi}{\partial \eta} \hat{e}_\eta = -\frac{1}{h_\xi} \frac{\partial \Phi}{\partial \xi} \hat{e}_\xi = -\frac{\cosh \xi - \cos \eta}{a} \frac{V}{\xi_2 - \xi_1} \hat{e}_\xi.$$

Thus, the electric field points in the $\hat{e}_\xi$ direction, which is perpendicular to the cylindrical surfaces, as expected for equipotential conductors. The surface charge densities on the facing surfaces of the two conductors are equal in magnitude but opposite in sign, and are given by

$$\sigma_1(\eta) = \varepsilon_0 \hat{e}_\xi \cdot \mathbf{E} = -\frac{\cosh \xi - \cos \eta}{a} \frac{\varepsilon_0 V}{\xi_2 - \xi_1}, \quad \sigma_2(\eta) = -\varepsilon_0 \hat{e}_\xi \cdot \mathbf{E} = \frac{\cosh \xi - \cos \eta}{a} \frac{\varepsilon_0 V}{\xi_2 - \xi_1}.$$

Here, $\sigma_1(\eta)$ and $\sigma_2(\eta)$ vary along the surfaces with the η coordinate, reflecting the nonuniform distribution of charge due to the cylindrical geometry. The opposite signs ensure that the total charge on one cylinder is balanced by the opposite charge on the other. The total charge per unit length on each cylinder can be calculated by integrating the surface charge density along the cylindrical surface. The line element is related to the bipolar scale factor h_η as $dl = h_\eta d\eta$. Therefore,

$$Q_1 = \int \sigma_1(\eta) dl = \int_0^{2\pi} \sigma_1(\eta) h_\eta d\eta = -\frac{2\pi\epsilon_0 V}{\xi_2 - \xi_1}, \quad Q_2 = -Q_1 = \frac{2\pi\epsilon_0 V}{\xi_2 - \xi_1}.$$

Finally, the capacitance per unit length is

$$C = \frac{2\pi\epsilon_0}{\xi_2 - \xi_1}.$$

We can also express the above result in terms of the cylindrical radii R_1 , R_2 , and their separation d . Notice that

$$\begin{aligned} \cosh(\xi_2 - \xi_1) &= \cosh \xi_2 \cosh \xi_1 - \sinh \xi_2 \sinh \xi_1 \\ &= \sqrt{1 + \frac{a^2}{R_2^2}} \sqrt{1 + \frac{a^2}{R_1^2}} - \phi_1 \frac{a}{R_2} \frac{a}{R_1} \\ &= \frac{1}{R_1 R_2} \left[\sqrt{a^2 + R_1^2} \sqrt{a^2 + R_2^2} - \phi_1 a^2 \right], \end{aligned}$$

and

$$d^2 = R_1^2 + R_2^2 + 2a^2 - 2\phi_1 \sqrt{a^2 + R_1^2} \sqrt{a^2 + R_2^2},$$

we have

$$\sqrt{a^2 + R_1^2} \sqrt{a^2 + R_2^2} = \frac{1}{2} \phi_1 (R_1^2 + R_2^2 + 2a^2 - d^2).$$

Then,

$$\cosh(\xi_2 - \xi_1) = \phi_1 \frac{R_1^2 + R_2^2 - d^2}{2R_1 R_2},$$

and the capacitance becomes

$$C = \frac{2\pi\epsilon_0}{\operatorname{arccosh} \left(\phi_1 \frac{R_1^2 + R_2^2 - d^2}{2R_1 R_2} \right)}.$$

4 Green Function

Following Jackson, the Green function is defined as the solution to

$$\nabla^2 G(\mathbf{x}, \mathbf{x}') = -4\pi \delta(\mathbf{x} - \mathbf{x}').$$

This Green function satisfies the appropriate boundary conditions on the cylinder surfaces and serves as a fundamental building block for solving more general electrostatic problems in this geometry.

In bipolar coordinates, the Dirac delta function is given by

$$\delta(\mathbf{x} - \mathbf{x}') = \frac{1}{h_\xi} \delta(\xi - \xi') \cdot \frac{1}{h_\eta} \delta(\eta - \eta') = \frac{(\cosh \xi - \cos \eta)^2}{a^2} \delta(\xi - \xi') \delta(\eta - \eta'),$$

while the Laplacian operator is

$$\nabla^2 = \frac{(\cosh \xi - \cos \eta)^2}{a^2} \left(\frac{\partial^2}{\partial \xi^2} + \frac{\partial^2}{\partial \eta^2} \right).$$

Substituting these expressions into the defining equation for the Green function, we obtain

$$\left(\frac{\partial^2}{\partial \xi^2} + \frac{\partial^2}{\partial \eta^2} \right) G(\xi, \eta; \xi', \eta') = -4\pi \delta(\xi - \xi') \delta(\eta - \eta').$$

The delta function in η parameter can be expressed in the following familiar form,

$$\delta(\eta - \eta') = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} e^{in(\eta - \eta')}.$$

Accordingly, the Green function can be expanded as

$$G(\xi, \eta; \xi', \eta') = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} g_n(\xi, \xi') e^{in(\eta - \eta')},$$

where the coefficient functions $g_n(\xi, \xi')$ depend only on ξ and must satisfy the differential equation

$$\left(\frac{\partial^2}{\partial \xi^2} - n^2 \right) g_n(\xi, \xi') = -4\pi \delta(\xi - \xi').$$

Excluding the delta function, the general solution to the differential equation for $n \neq 0$ is a linear combination of $e^{\pm n\xi}$. By imposing the Dirichlet boundary condition and requiring that the Green function vanishes on the cylindrical surfaces at $\xi = \xi_1$ and $\xi = \xi_2$, the coefficient functions $g_n(\xi, \xi')$ can be symmetrically constructed as

$$g_n(\xi, \xi') = A_n \sinh[n(\xi_{>} - \xi_2)] \sinh[n(\xi_{<} - \xi_1)],$$

where $\xi_{>}$ ($\xi_{<}$) is the larger (smaller) of ξ and ξ' . The constant A_n can be determined by matching the discontinuity in the derivative at $\xi = \xi'$ due to the delta function,

$$A_n = -\frac{4\pi}{n \sinh[n(\xi_2 - \xi_1)]}$$

Similarly, for $n = 0$, we have

$$g_0(\xi, \xi') = A_0(\xi_{>} - \xi_2)(\xi_{<} - \xi_1),$$

where

$$A_0 = -\frac{4\pi}{\xi_2 - \xi_1}.$$

Putting everything together, we will obtain the Green function

$$G(\xi, \eta; \xi', \eta') = \frac{2(\xi_2 - \xi_{>})(\xi_{<} - \xi_1)}{\xi_2 - \xi_1} + 2 \sum_{n \neq 0} \frac{\sinh[n(\xi_2 - \xi_{>})] \sinh[n(\xi_{<} - \xi_1)]}{n \sinh[n(\xi_2 - \xi_1)]} e^{in(\eta - \eta')}.$$

This agrees with Heyda [3]

References

- [1] J.D. Jackson, *Classical Electrodynamics*, 3rd ed. (John Wiley & Sons, 1999), Problem 2.8.
- [2] L. Landau and E. Lifshitz, *Electrodynamics of Continuous Media*, 2nd revised ed. (Butterworth-Heinemann Ltd., 1984), Problem 3.7.
- [3] J. Heyda, *A Green's function solution for the case of laminar incompressible flow between non-concentric circular cylinders*, Journal of Franklin Inst. 267, 25-34 (1959).